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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech. (Electric Vehicle Technology) (CBCS) Degree Examinations

May - June 2024

22EVT211 - BATTERY MANAGEMENT SYSTEM

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	With a neat diagram illustrate the construction and working principles of Lithium-ion battery.	8	L*2	2	1,2
b)	Explain Battery Management System inside a high-end shaver.	8	L2	1	1,2
c)	Outline the role of BMS in electric vehicle.	4	L1	1	1,2
2. a)	With necessary waveforms, elaborate role of DC-DC converter in BMS.	8	L2	1	1,2
b)	Derive the Gibbs free energy equation from the fundamentals of thermodynamics.	8	L2	2	1,2
c)	Write a note on Double layer capacitance occurring in battery electrodes.	4	L1	2	1,2
3. a)	Using fundamentals of chemical reaction elaborate kinetic and diffusion over potentials of battery.	8	L2	1	1,2
b)	Discuss the functionalities of power module used in BMS.	8	L2	2	1,2
c)	Write a note on Battery characteristics.	4	L1	1	1,2
Unit – II					
4. a)	Elaborate temperature measurement methods and uncertainty in measurements faced by BMS.	8	L2	3	1,3
b)	Elaborate necessity of cell balance and explain flying capacitance type cell balancing.	8	L2	4	1,3
c)	Explain Kelvin connection used for current measurement.	4	L1	3	1,3
5. a)	Explain SOC estimation technique based on Kalman filter.	8	L2	4	1,3
b)	Compare various modes of battery charging techniques.	8	L2	3	1,3
c)	Write a note on measurement uncertainty and its effect on BMS performance.	4	L1	4	1,3
6. a)	Elaborate various current sensors used for current measurements in BMS.	8	L2	3	1,3
b)	Elaborate passive cell balancing technique used in BMS.	8	L2	4	1,3
c)	Write a note on thermal management of battery pack.	4	L1	4	1,3
Unit – III					
7. a)	Explain the predictive State of Health model and role of BMS to implement the model.	8	L2	5	1,2
b)	Explain effect of over current fault and action taken by BMS to mitigate the same.	8	L1	5	1,2
c)	Write a note on Dual-Loop System with respect to BMS.	4	L1	5	1,2
8. a)	Explain Overcurrent excessive self-discharge and battery imbalance faults occurring in the battery.	8	L2	5	1,2
b)	Elaborate active method of impedance detection used in SOH estimation.	8	L2	5	1,2
c)	Summarize on excessive Capacity loss in battery pack.	4	L1	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
